Diagnosing non-identifiability in missing transverse momentum interpretations using CMS Open Data

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Abstract

Inclusive validation tests of collider observables can admit multiple nuisance explanations when analyses rely on reduced-observability data formats. This study investigates this identifiability problem using the missing transverse momentum (MET) distribution in $Z \to \mu^+\mu^-$ events reconstructed from CMS Open Data. Within the CMS Collaboration, MET performance has been extensively characterized using full detector information and dedicated systematic variations. By contrast, external analyses using CMS Open Data are necessarily restricted to reduced-tier representations, such as NanoAOD, which expose MET primarily as a reconstructed end-product. Distinct nuisance mechanisms are shown to reproduce indistinguishable inclusive MET distributions, demonstrating that compression of high-dimensional collider events into reduced observables can induce non-identifiability in standard goodness-of-fit validation.

The goal is not to improve MET reconstruction or reproduce collaboration-grade systematics, but to assess whether distinct, physically plausible nuisance explanations can be discriminated using only public-tier observables. Two low-dimensional phenomenological nuisance model families are constructed, correlated respectively with event topology (jet multiplicity reweighting) and recoil response (pT(Z)-correlated MET scale), and each is tuned to reproduce the observed MET distribution. A con-

trolled injection framework is used to evaluate whether these incompatible explanations can be tuned to reproduce the same target MET distribution and whether the inclusive projection is sufficient to discriminate between them. Stratified diagnostics conditioned on pT(Z) and jet multiplicity are further examined for their ability to restore discrimination.

The inclusive MET projection is found to provide insufficient discrimination across a range of physically motivated perturbations. A topology-mixture nuisance family reproduces the inclusive distribution with $\chi^2/\mathrm{dof} = 3.13/11$ ($p = 0.989$), while a detector-level perturbation derived from CMS unclustered-energy systematic variations lies near the boundary of inclusive discriminability ($\chi^2 = 18.28$, $p = 0.075$). Stratified projections conditioned on pT(Z) and jet multiplicity recover discrimination, yielding $\Delta\chi^2 = 1{,}432$ in the highest pT(Z) stratum and $\Delta\chi^2 = 3{,}415$ in the ≥2-jet category. These results do not challenge collaboration MET performance studies, but instead quantify an inference boundary relevant to external analyses using open data. The strength of the stratified discrepancy depends on the nuisance mechanism, reflecting which latent assumptions of the model are violated. These results reflect a general consequence of information compression in collider data tiers: reducing observable dimensionality to high-level reconstructed quantities removes the latent structure required to distinguish mechanistically distinct explanations at the inclusive level. The analysis demonstrates that conditional stratification provides a simple diagnostic for detecting nuisance-model degeneracies and mapping the identifiability boundary of compressed collider data representations.

1. Introduction

Analyses performed on compressed or reduced-tier datasets face an intrinsic statistical challenge: distinct physical mechanisms can produce indistinguishable predictions for marginal observables when latent structure is integrated out. In such settings, agreement between model and data in a one-dimensional validation distribution does not guarantee that the underlying nuisance mechanism is uniquely identified. This problem becomes particularly relevant for external analyses using publicly released collider datasets, where the available observables are intentionally compressed relative to the full detector reconstruction.

Many collider measurements rely on reduced analysis tiers, where high-dimensional detector information is compressed into summary observables such as missing transverse momentum (MET). While this reduction enables efficient analysis and public data release, it can introduce an identifiability limitation: distinct physical or instrumental mechanisms may produce indistinguishable distributions once events are projected into a compressed observable space. In statistical terms, the reduction $R(X) \to Y$ can induce degeneracy where $P(Y \mid M_1) \approx P(Y \mid M_2)$ despite $P(X \mid M_1) \neq P(X \mid M_2)$. However, partial recovery of identifiability may still be possible when conditioning on structural variables that survive the reduction, such as hadronic recoil or jet multiplicity. This work investigates such compression-induced degeneracy in missing transverse momentum inference using CMS Open Data and demonstrates that mechanisms indistinguishable in inclusive MET distributions become separable once conditional structure is examined. Such effects are particularly relevant for analyses performed on compressed public data tiers, where access to full detector-level information is unavailable and inference must proceed using reduced observables alone.

In the broader statistics literature this situation corresponds to an identifiability problem, in which distinct latent mechanisms produce indistinguishable predictions once high-dimensional structure is compressed into low-dimensional summary observables. Modern collider analyses normally mitigate such degeneracies using simultaneous multi-bin profile likelihood fits across orthogonal control regions, but these techniques rely on access to the full observable space and collaboration-internal systematic variations that are not available in reduced-tier public datasets.

Missing transverse momentum (MET) provides a particularly useful observable for examining this issue. MET is a highly composite quantity, constructed from the vector sum of reconstructed particle-flow candidates and subject to corrections that depend on jet calibration, pileup mitigation, and detector response. Its high-value tail is often used as a diagnostic of detector performance. At the same time, the NanoAOD format used in CMS Open Data exposes MET primarily as a reconstructed end-product rather than as a quantity derived from constituent detector information. This makes MET an ideal setting for studying how reduced observability affects the identifiability of nuisance mechanisms. Within the CMS Collaboration, MET reconstruction and performance have been studied in detail using full detector information, dedicated calibration constants, and systematic uncertainty variations, making extensive use of

control samples such as Z→ll and γ+jet events to characterize MET scale, resolution, and anomalous behavior.

In parallel, CMS has released a substantial fraction of its LHC data to the public, enabling external researchers to perform analyses using well-defined reduced data tiers such as NanoAOD. These releases have supported a growing body of peer-reviewed work, including studies of jet substructure, dimuon spectra, and methodological analyses based on public data. However, the reduced-tier formats provided for open data necessarily compress detector-level information and expose high-level observables, such as MET, primarily as reconstructed end-products. While this design greatly improves accessibility, it also limits the range of detector and reconstruction effects that can be explicitly modeled or varied by external analysts.

As a consequence, external analyses using CMS Open Data face a distinct interpretive challenge: even when agreement between data and simulation is observed for standard validation distributions, it is not always clear whether the underlying causal mechanisms responsible for that agreement are uniquely determined by the available observables. In other words, agreement does not automatically imply identifiability. This distinction is particularly relevant for MET, where multiple detector-related effects can plausibly contribute to similar distributional features.

The possibility that distinct nuisance mechanisms can produce indistinguishable predictions for marginal observables is well recognised in collider statistics. Contemporary LHC analyses typically address such degeneracies using multidimensional control regions and simultaneous profile-likelihood fits across correlated observables, often supplemented by auxiliary calibration measurements [12]. More recently, simulation-based inference and related likelihood-learning techniques have explored similar identifiability questions in high-dimensional observable spaces [13, 14]. These approaches implicitly rely on access to rich detector-level information and extensive sets of systematic-variation branches available within collaboration analysis frameworks. By contrast, external analyses using CMS Open Data operate on compressed representations such as NanoAOD, where many of the underlying degrees of freedom are no longer directly accessible. The present work therefore addresses a complementary question: to what extent competing nuisance explanations remain distinguishable when inference is restricted to the reduced observable space available at the public data tier.

Reduced collider observables compress high-dimensional event structure into a limited set of summary statistics; this compression can erase the information required to distinguish physically distinct nuisance mechanisms, creating non-identifiability in inclusive validation tests.

The present study addresses this issue by focusing on $Z \to \mu^+\mu^-$ events, which provide a well-established standard candle for MET studies. In these events, genuine MET is expected to be negligible, such that the observed MET distribution is dominated by instrumental effects. The analysis is restricted to publicly available NanoAOD-level observables, explicitly adopting the perspective of an external analyst operating within the constraints of CMS Open Data.

Importantly, the degeneracy demonstrated here is constructed using real collider events from CMS Open Data rather than purely simulated examples. This allows the analysis to expose inference structure that arises in practical collider datasets rather than in idealized Monte Carlo studies.

Rather than attempting to reproduce collaboration-grade MET systematics or to propose improvements to MET reconstruction, the objective is to assess the discriminating power of public-tier observables with respect to competing, physically plausible nuisance explanations for MET tails. Two phenomenological nuisance families are examined, representing distinct mechanisms by which the inclusive MET distribution may be distorted: a recoil-response scale distortion correlated with pT(Z), and a topology-mixture reweighting that alters the relative population of jet-multiplicity categories. These mechanisms can produce similar distortions in the marginal MET distribution but imply different conditional structure when the data are stratified by recoil or by event topology.

Using a controlled injection framework and a χ^2 fit criterion, the analysis evaluates whether these nuisance families can be tuned to reproduce the same target MET distribution and whether the inclusive projection is sufficient to discriminate between them. The effectiveness of stratified diagnostics — conditioned on pT(Z) and jet multiplicity, both available in NanoAOD — is further examined for their ability to break the inclusive degeneracy.

By quantifying when and how such degeneracies persist, this work aims to delineate an inference boundary relevant to external analyses of CMS Open Data. The results

are not intended to challenge the validity of CMS MET performance studies, but rather to clarify what kinds of mechanistic conclusions are and are not supported by reduced-tier public representations. More broadly, the approach provides a template for assessing identifiability limits of high-level observables in open collider datasets. The goal of this study is therefore not to propose an improved model of MET, but to map how different nuisance mechanisms project onto compressed observables and to identify simple diagnostics that restore mechanistic discrimination.

2. CMS Open Data and Analysis Scope

The analysis presented in this work is based exclusively on publicly released CMS Open Data corresponding to proton–proton collisions recorded in 2016 at a center-of-mass energy of 13 TeV by the CMS detector. The CMS Open Data program provides external access to research-grade datasets, associated Monte Carlo simulations, and documentation intended to enable reproducible analyses outside the collaboration.

2.1 Open Data Tiers and Design Philosophy

CMS Open Data are released in multiple data tiers that differ in information content, file size, and intended use. For Run 2 data, the NanoAOD format represents the most compact tier designed for physics analysis, containing high-level reconstructed objects and derived quantities in a flat ntuple structure. NanoAOD is explicitly optimized for accessibility and computational efficiency, enabling analyses to be performed without reliance on the full CMS software stack or detector-level reconstruction code.

This design necessarily entails a loss of low-level information. In particular, NanoAOD does not provide access to individual particle-flow candidates, detailed reconstruction intermediates, or the full sequence of corrections and calibrations applied internally within CMS. As a result, certain detector effects—especially those associated with unclustered energy, rare reconstruction pathologies, or detailed pileup mitigation algorithms—cannot be independently recomputed or varied by external analysts.

The present work explicitly embraces these constraints. Rather than treating reduced observability as a limitation to be overcome, such compression is treated as a defining feature of the inference problem under study.

2.2 MET Representation in NanoAOD

In NanoAOD, missing transverse momentum is provided as a reconstructed object (`MET_pt`, `MET_phi`) that reflects the final CMS MET definition after standard corrections and pileup mitigation procedures. While jet kinematics, muon kinematics, and basic pileup proxies (e.g., number of reconstructed primary vertices) are available, the internal composition of MET—including contributions from unclustered energy—is not exposed.

Consequently, MET in NanoAOD should be understood as an end-product observable rather than a quantity that can be re-derived from constituent objects. This has important implications for causal interpretation: different detector- or reconstruction-level mechanisms may map onto similar MET distributions without being distinguishable at the public tier.

The analysis in this work does not attempt to reverse-engineer or reinterpret the CMS MET reconstruction. Instead, MET is treated as an observable whose distributional behavior is to be compared between data and simulation under phenomenological perturbations applied at the level of available observables.

2.3 Scope of Simulation Use

Monte Carlo simulation samples corresponding to Z+jets processes are used as a reference model for comparison with data. Consistent with the public-data context, only the simulation products provided through the CMS Open Data portal are used, without access to internal systematic variations or detector recalibration tools.

Nuisance models introduced in this work are therefore applied at the level of the simulated MET observable or via correlations with other available high-level quantities. These models are not intended to reproduce CMS systematic uncertainty procedures, but to represent distinct, physically plausible explanations that an external analyst could reasonably entertain given the accessible information.

2.4 Deliberate Scope Limitations

For clarity, the following scope restrictions apply:

- This work does not attempt to validate or supersede CMS MET performance studies.

- No detector-level recalibration or re-reconstruction is performed.
- No claims are made regarding optimal MET modeling or correction strategies.
- All conclusions are conditional on the information available at the NanoAOD tier.

Within this deliberately constrained scope, the analysis addresses a narrower but well-defined question: to what extent can distinct instrumental explanations for MET tails be discriminated using public-tier observables alone?

3. Event Selection

3.1 Input observables

The selection and baseline analysis read the following NanoAOD branches from the `Events` tree:

- **Muon kinematics and identity:** `Muon_pt`, `Muon_eta`, `Muon_phi`, `Muon_mass`, `Muon_charge`
- **Muon quality:** `Muon_tightId` (boolean), `Muon_pfRelIso04_all` (PF relative isolation, cone $\Delta R = 0.4$)
- **Missing transverse momentum:** `MET_pt`
- **Jets:** `Jet_pt`, `Jet_eta`, `Jet_phi`
- **Event count:** `nMuon`

No lower-level detector quantities are used. The analysis operates entirely at the NanoAOD reconstructed-object tier. The muon identification branch is resolved at runtime in order of preference: `Muon_tightId`, `Muon_mediumId`, `Muon_looseId`; the isolation branch is resolved in order: `Muon_pfRelIso04_all`, `Muon_pfRelIso03_all`, `Muon_miniPFRelIso_all`. For the dataset used in this analysis, `Muon_tightId` and `Muon_pfRelIso04_all` are present and are used throughout.

3.2 Muon selection and dimuon construction

Per-muon quality cuts are applied before any event-level selection. A muon is retained if it satisfies all of the following requirements simultaneously:

- p_T > 15 GeV
- $|\eta| < 2.4$
- `Muon_tightId` = 1 (CMS Tight muon identification)

- `Muon_pfRelIso04_all` < 0.15 (PF relative isolation, $\Delta R = 0.4$ cone)

Both muons are required to satisfy p*_T* > 15 GeV and |η| < 2.4. In addition, the leading muon in the selected pair must satisfy p*_T* ≥ 25 GeV. All muons passing the per-muon cuts are retained as candidates. No multiplicity requirement is applied at this stage. Four-momentum vectors are constructed from (`Muon_pt`, `Muon_eta`, `Muon_phi`, `Muon_mass`) using the `vector` library Momentum4D representation.

Dimuon pairs are formed by taking all unique two-muon combinations within each event and retaining only opposite-sign pairs. For events containing multiple quality-selected muons, all opposite-sign combinations are evaluated. The invariant mass of each OS pair is computed from the four-vector sum.

If multiple opposite-sign muon pairs satisfy the baseline selection, the Z candidate is defined as the pair whose invariant mass is closest to the nominal Z boson mass (91.1876 GeV). This "best pair" criterion is applied event by event without iterating over alternative selections. The leading muon p*_T* of the best pair — defined as the higher of the two muon transverse momenta — must satisfy p*_T*(μ_1) ≥ 25 GeV. Events in which no OS pair exists, or in which the best-pair leading muon fails this threshold, are rejected.

3.3 Z-candidate definition

The Z-candidate mass is taken from the four-vector sum of the best OS pair selected in Section 3.2. A mass window requirement of 80 GeV ≤ $m(\mu^+\mu^-)$ ≤ 100 GeV is applied as a hard selection cut. Events whose best-pair mass falls outside this window are excluded from the analysis sample. This window defines the $Z \to \mu^+\mu^-$ signal region used in all downstream inclusive and stratified comparisons.

The dimuon transverse momentum p*_T*(Z) is taken as the magnitude of the transverse component of the four-vector sum of the two selected muons. It serves as the primary recoil variable for the pT(Z) stratification in Section 6.2.

A separate histogram-mode estimator in the narrower window 85–97 GeV is used in the Phase 0 feasibility check to confirm the Z peak position without fitting assumptions. This narrower window is a diagnostic convention and does not define the analysis sample.

3.4 Jet cleaning and jet-multiplicity definition

Hadronic activity is quantified by counting jets reconstructed by the CMS particle-flow algorithm. A jet is included in the multiplicity count if it satisfies:

- p*_T*(jet) > 30 GeV
- ΔR(jet, μ_1) > 0.4 **and** ΔR(jet, μ_2) > 0.4

where μ_1 and μ_2 are the two muons forming the selected Z candidate, and $\Delta R = \sqrt{(\Delta\eta^2 + \Delta\varphi^2)}$. Jets that fail either cleaning condition are removed from the count before applying the p*_T* threshold.

This cleaning step prevents the particle-flow jet algorithm from counting the selected muons as jets. Any jet within a cone of $\Delta R < 0.4$ around either Z-candidate muon is removed, ensuring that the jet multiplicity reflects genuine hadronic recoil activity rather than jet-muon object overlap.

Events are assigned to one of three exclusive jet-multiplicity categories:

- **0-jet:** no jets with p*_T* > 30 GeV survive the cleaning requirement
- **1-jet:** exactly one such jet survives
- **≥2-jet:** two or more such jets survive

These three categories are used for the jet-multiplicity stratification in Section 6.3.

3.5 Analysis sample used in Section 6

The selection described in Sections 3.2–3.4 was applied to the full CMS Open Data DoubleMuon 2016 NanoAOD dataset. A total of 2,147,195 events were processed. After applying all muon quality cuts, the opposite-sign requirement, the leading muon p*_T* threshold of 25 GeV, and the Z-mass window of 80–100 GeV, 233,524 events were retained. The overall selection efficiency is approximately 10.9%.

This sample of 233,524 $Z \to \mu^+\mu^-$ candidates constitutes the analysis sample for all comparisons in Section 6. The same selected events are used for the inclusive MET comparison (Section 6.1), the pT(Z)-stratified comparisons (Section 6.2), the jet-multiplicity-stratified comparisons (Section 6.3), and the MET binning robustness checks (Section 6.4). No additional event-level requirements are applied within the results section; the stratification into sub-samples uses the dimuon p*_T* and

jet-multiplicity variables already computed during the selection. This selected sample is used for all inclusive and stratified comparisons reported in Section 6.

4. Nuisance Model Families

4.1 Motivation for phenomenological nuisance families

The nuisance models considered here are intentionally low-dimensional phenomenological parameterisations. Their purpose is not to reproduce collaboration-grade detector systematic models, but to represent distinct classes of physical explanations for distortions in the missing transverse momentum distribution that an external analyst might entertain when working with CMS Open Data. The analysis evaluates whether such incompatible mechanisms can produce indistinguishable agreement with the inclusive observable when only reduced-tier information is available.

Two families are considered, each parameterised by a single continuous scalar. Low dimensionality is deliberate: excessive model flexibility would trivially explain any distributional feature, making non-identifiability uninformative. A single-parameter family imposes a strong constraint on the allowed distortion shape; degeneracy under such a constraint is therefore a meaningful result.

The two families are mutually incompatible in mechanism: one alters the MET magnitude of individual events, the other alters the population weights across event topologies. Agreement on the inclusive distribution is therefore not tautological but reflects a genuine degeneracy of the marginal projection.

4.2 Fitting strategy

The identifiability test is implemented as a controlled injection. Family A is applied to the nominal MET distribution at a known working point α_0, producing a target distribution that represents a specific, calibrated departure from the nominal. Family B is then independently fitted to the same target by grid search over its parameter β. Fit quality is quantified using the χ^2 statistic:

```
χ² = Σ_k (h_k^model − h_k^target)² / (h_k^target + 1)
```

where h_k denotes the bin count in histogram bin k, and the (obs + 1) denominator provides stability in bins with low target occupancy. Degrees of freedom for Family B

are n_bins − 1 fitted parameter.

Inclusive fit quality determines the working point. For a given α_0, if Family B achieves an acceptable p-value on the inclusive χ^2, the two families are non-identifiable on the inclusive projection. Stratified comparisons are then applied at the inclusive best-fit parameters, without per-stratum refitting, to test whether conditional projections restore discrimination.

The simplified χ^2 goodness-of-fit criterion should be understood as a diagnostic tool for comparing nuisance mechanisms rather than as a full profile-likelihood inference in the standard LHC sense [12]. The goal is not optimal parameter estimation but the detection of inclusive degeneracy and its resolution through conditional projections.

4.3 Family A — Recoil-Response Distortion

Family A represents the hypothesis that MET tails arise from a miscalibrated hadronic recoil response. In $Z \to \mu^+\mu^-$ events, the Z boson recoils against hadronic activity, and the magnitude of the MET reflects the vector imbalance of that recoil. A fractional miscalibration in the hadronic response would shift the reconstructed MET proportionally to the recoil magnitude, which is directly approximated by pT(Z), the transverse momentum of the dimuon system.

The model modifies the event-level MET as:

```
MET'_i = MET_i × (1 + α × clip(pT(Z)_i, 0, 100) / 100)
```

where pT(Z)_i is the dimuon transverse momentum in GeV, clip(·, 0, 100) caps the effective recoil scale at 100 GeV to prevent unbounded scaling in the high-recoil tail, and α is the distortion amplitude. Positive α inflates MET; negative α deflates it.

The parameter search range is $\alpha \in [-0.30, +0.30]$ in steps of 0.01. The injection working point used throughout this analysis is $\alpha_0 = 0.05$, corresponding to a 5% maximum MET scale distortion in the highest-recoil events, scaling linearly with pT(Z) below 100 GeV. Family A leaves the relative population of jet-multiplicity categories unchanged.

4.4 Family B — Topology-Mixture Reweighting

Family B represents the hypothesis that MET tails arise from a shift in the event-topology composition of the sample rather than from a miscalibration of the MET scale within any given topology. Such distortions may arise from mismodelling of jet production rates, residual efficiency corrections that vary across jet categories, or other effects that alter the effective mixture of zero-jet, one-jet, and multi-jet events contributing to the inclusive distribution.

The model applies per-event weights according to jet multiplicity:

Jet category	Weight
0 jets	1
1 jet	$1 + \beta$
$\geq$2 jets	$1 + 2\beta$

Jet multiplicity is counted using the same cleaned jet definition as in the event selection ($p_T > 30$ GeV, $\Delta R > 0.4$ from both Z-candidate muons; Section 3.4). The weights are normalised so that the total weighted event count equals the unweighted count, preserving the inclusive sample size.

The parameter β controls the degree to which jet-rich topologies are upweighted relative to zero-jet events. The search range is $\beta \in [-0.40, +0.40]$ in steps of 0.01. At $\beta > 0$, events with more jets are upweighted, shifting the inclusive MET distribution toward higher values because multi-jet events have systematically larger MET than zero-jet events.

Crucially, Family B does not alter the MET value of any individual event. It changes only the effective population of each jet-multiplicity stratum.

The nuisance families are chosen to represent qualitatively distinct physical mechanisms: topology mixture modifies the composition of event kinematics entering the MET distribution, while recoil-scale perturbations model detector-level response distortions. Demonstrating degeneracy across these mechanism classes indicates that the indistinguishability arises from observable compression rather than from trivial functional parameterizations.

4.5 Mechanistic Contrast

The two families modify the inclusive MET distribution through fundamentally different mechanisms:

	Family A	Family B
Mechanism	Recoil-response distortion	Topology-mixture reweighting
Parameter	α	β
Effect on event-level MET	Scales MET with pT(Z)	None
Effect on jet-category populations	None	Upweights multi-jet events
Injection working point	α = 0.05	β = +0.290 (best fit)

Family A is sensitive to the recoil–MET relationship and is exposed by conditioning on pT(Z). Family B is sensitive to population consistency across jet categories and is exposed by conditioning on jet multiplicity. Neither diagnostic is implicit in the inclusive MET distribution, which is what makes the inclusive degeneracy possible.

4.6 Scope of Nuisance Parameterisations

The nuisance models used in this study are intentionally simplified representations of broader classes of detector and modelling effects. The goal is not to reproduce collaboration-grade systematic uncertainties, which require detailed detector simulation and calibration studies, but to construct phenomenological families that capture distinct mechanisms by which the inclusive MET distribution may be distorted. The analysis therefore tests identifiability at the level of mechanism rather than at the level of detailed detector modelling.

If two mechanistically distinct nuisance explanations cannot be distinguished in the inclusive projection even in this simplified setting, the ambiguity will persist — and likely worsen — for more realistic high-dimensional systematic models. The simplified parameterisations therefore provide a conservative lower bound on the identifiability challenge: demonstrating degeneracy here implies that more complex nuisance families face at least as severe an identifiability constraint. Because the nuisance families

are restricted to a single parameter, the degeneracy demonstrated in the inclusive projection represents a conservative lower bound on the identifiability challenge. Increasing the dimensionality of the nuisance parameterisation without introducing additional observables generally enlarges the space of distributions compatible with the data rather than reducing it.

5. Identifiability Test Design

This section describes the statistical framework used to test whether the two nuisance families defined in Section 4 can be distinguished using the inclusive MET distribution, and how conditional projections are used to recover discriminating information absent from the inclusive view. The analysis uses the full selected sample of 233,524 $Z \to \mu^+\mu^-$ events described in Section 3; no train/test split is applied.

5.0 Identifiability under compressed observables

A nuisance model θ is said to be identifiable from an observable Y if no two distinct parameter values $\theta_1 \neq \theta_2$ produce the same distribution: $P(Y \mid \theta_1) = P(Y \mid \theta_2)$ implies $\theta_1 = \theta_2$. When Y is a compressed projection of a higher-dimensional observable X — as occurs when full detector-level data are reduced to a summary quantity such as MET — this condition can fail even when the generating mechanisms are well separated in the original space. Formally, the reduction map $R: X \to Y$ can induce degeneracy of the form $P(Y \mid \theta_1) \approx P(Y \mid \theta_2)$ despite $P(X \mid \theta_1) \neq P(X \mid \theta_2)$.

In the present analysis, X represents the full kinematic state of the event (including per-particle detector response, pileup multiplicity, and hadronic recoil structure), while Y = MET, the scalar missing transverse momentum exposed at the NanoAOD tier. The nuisance families A and B represent distinct mechanisms operating in the latent space X, yet the question posed here is whether their effects on the marginal distribution $P(Y \mid \theta)$ remain distinguishable after projection onto MET alone.

Non-identifiability in $P(Y \mid \theta)$ is not merely a statistical-power limitation: it reflects a structural property of the observable space. Collecting more events does not resolve a degeneracy when $P(Y \mid \theta_1) = P(Y \mid \theta_2)$ exactly; even the approximate case $P(Y \mid \theta_1) \approx P(Y \mid \theta_2)$ implies that the information required to distinguish θ_1 from θ_2 has been removed by the reduction R. Partial recovery of identifiability may still be achievable

by conditioning on structural variables that survive the reduction — such as pT(Z) or jet multiplicity, both available in NanoAOD — since the conditional distributions $P(Y \mid \theta, Z = z)$ may still satisfy $P(Y \mid \theta_1, Z = z) \neq P(Y \mid \theta_2, Z = z)$ even when the marginals are degenerate.

The identifiability landscape figure (Section 6) summarises the empirical result: the horizontal axis encodes the inclusive p-value (how well a family fits the target in the marginal MET projection), and the vertical axis encodes the maximum per-stratum $\Delta\chi^2$ (how strongly conditional projections discriminate the family from the injected truth). A family occupying the upper-right region of this diagram — high inclusive p-value combined with high max $\Delta\chi^2$ — represents the structurally hazardous case: inclusive validation suggests agreement, but conditional structure reveals that the mechanism is inconsistent with the truth. Family B occupies this region at the analysis working point.

5.1 Injection Framework

A target distribution is constructed by applying Family A (recoil-response distortion) to the observed MET values at a fixed injection parameter α_0:

```
MET_target(i) = MET(i) × (1 + α₀ × clip(pT(Z)_i, 0, 100) / 100)
```

This target represents the observed distribution as it would appear under a known, controlled nuisance distortion. Family A trivially recovers the target because it generated it; the nontrivial question is whether Family B (topology-mixture reweighting) can also reproduce the same target on the inclusive MET histogram, despite operating through a mechanistically different mechanism.

The injection framework has two advantages over a direct comparison of models to data. First, it provides a ground truth: the injected distortion amplitude α_0 is known exactly, so the quality of Family A's recovery is a calibration check. Second, it isolates the identifiability question from the absolute normalization of the data: both families are evaluated against the same target histogram, so differences in their fit quality reflect differences in shape-matching ability, not overall normalisation.

5.2 Inclusive Fit Criterion

Both families are fitted independently to the target histogram by grid search over their respective parameters. The fit metric is:

```
χ²(model, target) = Σ_k (h_k^model − h_k^target)² / (h_k^target + 1)
```

where the sum runs over all MET histogram bins k, h_k denotes the bin count, and the (h_k^target + 1) denominator provides numerical stability in bins with low target occupancy. This form is analogous to a Pearson χ^2 but uses the target bin count rather than the model prediction in the denominator, avoiding sensitivity to model parameters in the uncertainty estimate.

The MET histogram uses 12 variable-width bins spanning 0–200 GeV, with boundaries at [0, 5, 10, 15, 20, 30, 40, 50, 60, 80, 100, 150, 200] GeV. Finer binning at low MET (where most events lie) provides sensitivity to shape changes in the bulk of the distribution; coarser binning above 50 GeV reflects the sparse high-MET tail.

Family A is searched over $\alpha \in [-0.30, +0.30]$ in steps of 0.01. Family B is searched over $\beta \in [-0.40, +0.40]$ in steps of 0.01. The best-fit parameter for each family is the value minimising χ^2 over the grid.

A formal p-value for Family B is computed from the chi-squared survival function with dof = (number of bins) − 1 = 11 degrees of freedom, where the subtraction of one accounts for the single fitted parameter β. Family A, which generated the target distribution, achieves $\chi^2(A) = 0$ trivially and is not assigned a p-value. The working criterion for inclusive non-identifiability is $p(B) > 0.05$: if Family B cannot be rejected at the 5% level on the inclusive projection, the two families are operationally non-identifiable in that view.

5.3 Conditional Evaluation Without Refitting

After the inclusive fit, both families are evaluated in conditional subsamples defined by pT(Z) and jet multiplicity. Crucially, the nuisance parameters are held fixed at the values obtained from the inclusive fit; no per-stratum refitting is performed.

This design is deliberate. Per-stratum refitting would allow each family to optimise against the conditional distribution independently, potentially masking genuine differences in their internal structure. By holding parameters fixed, the stratified evalu-

ation tests whether the inclusive best-fit parameters also produce acceptable descriptions of the conditional distributions, without any additional degrees of freedom.

The per-stratum discrimination is quantified by:

```
Δχ²(stratum) = χ²(B, stratum) − χ²(A, stratum)
```

Because $\chi^2(A) \approx 0$ in every stratum (Family A generated the target distribution at the same parameter values used in the stratified evaluation), $\Delta\chi^2 \approx \chi^2(B)$ in each stratum. A large $\Delta\chi^2$ indicates that Family B's best-fit parameter, which was sufficient for the inclusive distribution, fails to reproduce the target when conditioned on the stratification variable.

The pT(Z) stratification uses four bins: [0, 10), [10, 30), [30, 60), and ≥60 GeV. The jet-multiplicity stratification uses three categories: 0-jet, 1-jet, and ≥2-jet, with the same jet definition as in the event selection (Section 3.4).

The conditional evaluation deliberately fixes nuisance parameters at the values obtained from the inclusive fit rather than allowing refitting within each stratum. This design converts the stratified comparison into a consistency test rather than a parameter-estimation procedure. Allowing parameters to refit independently in each stratum would permit each nuisance family to absorb conditional structure separately, potentially masking mechanistic inconsistencies between the model and the injected target. The fixed-parameter evaluation therefore functions analogously to a held-out diagnostic: the question being tested is whether the inclusive best-fit parameters remain consistent with the conditional distributions without additional degrees of freedom. The stratified evaluation therefore constitutes a held-out consistency test rather than a parameter-estimation step. Allowing the nuisance parameters to float independently in each conditional subset would introduce additional degrees of freedom and would test the flexibility of the parameterisation rather than the internal consistency of the mechanism inferred from the inclusive distribution.

5.4 Working-Point Selection

The injection amplitude α_0 determines the size of the distortion injected into the target distribution and therefore controls how far the target departs from the nominal MET distribution. At large α_0, the distortion is large enough that even the inclusive distribution distinguishes the two families; at small α_0, the distortion is undetectable

and neither family can be constrained.

To identify the practical degeneracy regime, the analysis was run at four injection amplitudes: $\alpha_0 \in \{0.02, 0.05, 0.10, 0.15\}$. The inclusive p-value for Family B as a function of injection amplitude provides a threshold map:

- At $\alpha_0 = 0.02$, both families fit the target with p(B) near unity.
- At $\alpha_0 = 0.05$, Family B achieves $\chi^2(B) = 3.13$, dof = 11, p = 0.989.
- At $\alpha_0 = 0.10$, Family B begins to fail the inclusive criterion ($\chi^2(B) \approx 41$).
- At $\alpha_0 = 0.15$, Family B is clearly rejected on the inclusive distribution.

The working point $\alpha_0 = 0.05$ is selected for the main demonstration because it lies firmly within the non-identifiable regime: at this amplitude, a 5% maximum MET scale distortion in the highest-recoil events is invisible to the inclusive projection despite a sample of more than 2×10^5 events, yet conditional projections recover clear discrimination. This choice represents a physically plausible distortion amplitude — small enough to pass inclusive validation, large enough to have interpretive consequences — rather than a worst-case or best-case scenario.

6. Results

After the selection described in Section 3, a total of 233,524 $Z \to \mu^+\mu^-$ candidate events remain from 2,147,195 processed events. All quantitative comparisons in this section use this selected sample. Because the analysis operates directly on reconstructed CMS Open Data events, the observed degeneracy reflects the behavior of realistic collider data products rather than properties of a specific simulation model.

The three nuisance families examined here occupy distinct positions along an identifiability spectrum: Family A represents the injected truth, Family B exhibits deep inclusive degeneracy, and Family C lies near the boundary of inclusive discriminability. Figure 5 provides a summary of this landscape at the end of this section.

The results are organised as follows. Section 6.1 presents the inclusive missing transverse momentum comparison, which establishes the working point for the identifiability demonstration. Section 6.2 examines the same comparison after conditioning on pT(Z), the primary recoil variable. Section 6.3 presents the jet-multiplicity stratification. Section 6.4 documents the robustness of the inclusive result to MET bin-

ning. Section 6.5 presents a physically grounded nuisance variation derived from CMS unclustered-energy systematic branches.

6.1 Inclusive MET degeneracy

The analysis sample consists of 233,524 $Z \to \mu^+\mu^-$ candidates selected from CMS Open Data (DoubleMuon, 2016). Both nuisance families were fitted independently to the inclusive MET histogram. Family A (recoil-response distortion) was used to construct the target distribution at the working point $\alpha = 0.05$, representing a 5% MET scale distortion growing linearly with min(pT(Z), 100 GeV)/100. Family B (topology-mixture reweighting) was then fitted to the same target by grid search over $\beta \in [-0.40, +0.40]$.

The best-fit parameters and inclusive fit quality are summarised in Table 1.

Table 1. Inclusive MET fit quality. The target distribution was constructed by applying Family A at the working point $\alpha = 0.05$. All families were fitted independently to this target using a χ^2 statistic with (obs + 1) denominator. Degrees of freedom are (number of histogram bins) − 1 fitted parameter. Family A, which generated the target, trivially achieves $\chi^2 = 0$.

Model	Best parameter	χ^2	dof	p-value
Family A	$\alpha = +0.050$	0.00	—	—
Family B	$\beta = +0.290$	3.13	11	0.989
Family C	$\gamma = +0.200$	18.28	11	0.075

Family B achieves $\chi^2(B) = 3.13$ on 11 degrees of freedom, corresponding to $p = 0.989$. The inclusive projection provides no statistical basis to reject the topology-mixture model at this working point. The result is shown visually in Fig. 1, which presents the inclusive MET distributions for the target, Family A, and Family B on a logarithmic scale, together with the ratio of each model to the target in the lower panel. The three distributions are visually indistinguishable across the bulk of the spectrum (MET < 100 GeV); a small excess from Family B appears in the sparse high-MET tail above 100 GeV, which accounts for the residual $\chi^2 = 3.13$.

These results demonstrate that the inclusive MET distribution, despite containing 233,524 selected events, admits a topology-mixture explanation with goodness of fit statistically equivalent to the injected recoil model.

As an empirical noise-floor check, the inclusive χ^2 statistic was recomputed on 1,000 bootstrap resamplings of the target MET distribution using the same 12-bin histogram and the same χ^2 metric. Each bootstrap sample draws 233,524 events with replacement from the target and is compared against the target histogram directly. The resulting null distribution has a median of $\chi^2 = 10.6$ and spans the range 2.2–29.8. The $\chi^2(B) = 3.13$ obtained for the topology-mixture family falls at the 1.2nd percentile of this distribution (Fig. 2), demonstrating that the inclusive agreement of Family B is not merely within the formal rejection threshold but is substantially better than the typical level of sampling variability intrinsic to the target histogram itself. The topology-mixture model, with its smooth optimised-weight histogram, provides a fit quality well inside the noise floor of the target distribution.

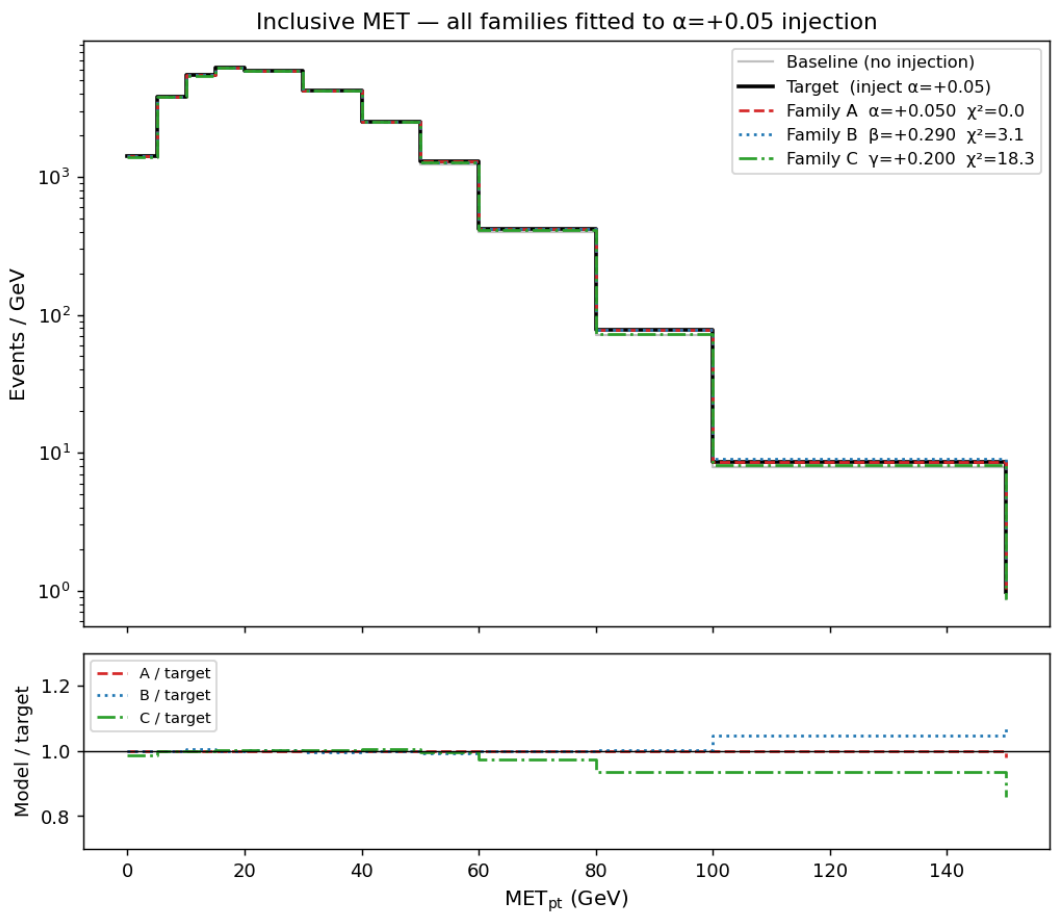

Figure 1: Inclusive MET distributions for the target (black), Family A (dashed red), Family B (dotted blue), and Family C (dash-dot green) on a logarithmic scale. The lower panel shows the ratio of each model to the target. Family B achieves $\chi^2(B) = 3.13$ (dof = 11, p = 0.989); Family C achieves $\chi^2(C) = 18.28$ (dof = 11, p = 0.075). All families are fitted to the inclusive histogram only. The sample comprises 233,524 $Z \to \mu^+\mu^-$ candidates from CMS Open Data (DoubleMuon 2016).

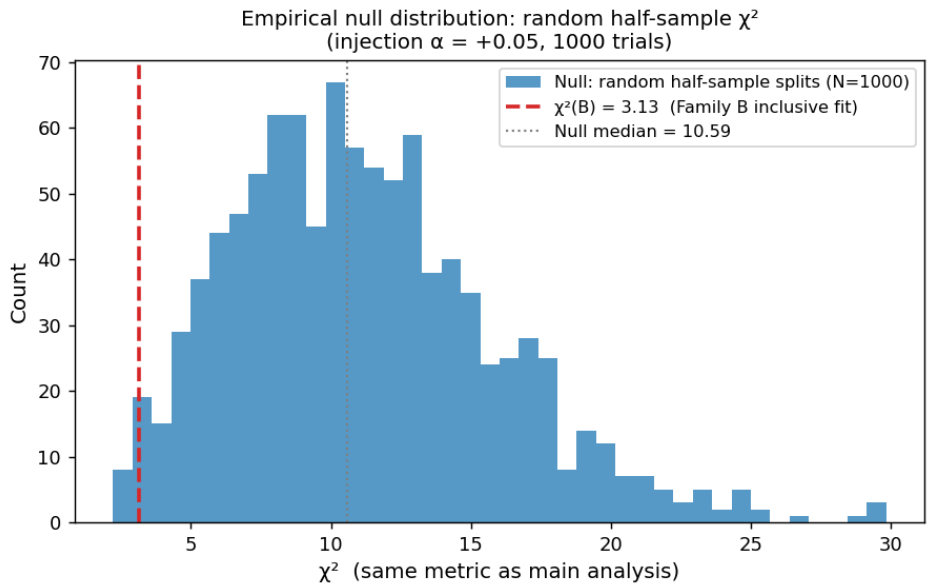

Figure 2: Empirical null distribution of the inclusive χ^2 statistic from 1,000 bootstrap resamplings of the target MET histogram (blue). Each bootstrap sample draws 233,524 events with replacement from the target. The dashed red line marks $\chi^2(B) = 3.13$, the inclusive fit for Family B. The null median is $\chi^2 = 10.6$ (dotted gray). Family B falls at the 1.2nd percentile of the null, confirming that the inclusive agreement is well within the sampling noise floor of the target distribution.

6.2 Recoil-stratified MET distributions

To probe whether the inclusive degeneracy conceals differences in the internal event structure, both models were evaluated—without refitting—in four pT(Z) strata following the protocol binning: [0, 10), [10, 30), [30, 60), and ≥60 GeV. The separation between the two nuisance families is quantified using $\Delta\chi^2 = \chi^2(B) - \chi^2(A)$, evaluated within each stratum while keeping the nuisance parameters fixed at the values obtained from the inclusive fit. Because Family A generated the target distribution, $\chi^2(A) \approx 0$ in each stratum and $\Delta\chi^2 \approx \chi^2(B)$.

Table 2. Per-stratum discrimination by recoil activity. $\Delta\chi^2 = \chi^2(B) - \chi^2(A)$ is evaluated in each pT(Z) bin using the best-fit parameters from the inclusive fit only. Both families were fitted to the inclusive distribution; the stratified comparison uses the inclusive best-fit parameters without refitting.

pT(Z) bin (GeV)	N	$\Delta\chi^2$
< 10	92,608	392
10 ≤ pT(Z) < 30	89,387	95

pT(Z) bin (GeV)	N	$\Delta\chi^2$
$30 \leq pT(Z) < 60$	34,312	513
≥ 60	17,217	1,432

The overall trend is that $\Delta\chi^2$ grows with recoil activity, with the largest separation in the highest pT(Z) bin ($\Delta\chi^2 = 1{,}432$) and the smallest in the 10–30 GeV bin ($\Delta\chi^2 = 95$). This pattern is consistent with the design of Family A, which produces MET distortions proportional to pT(Z): events with higher pT(Z) are shifted more, and Family B—which does not scale MET with pT(Z)—cannot reproduce that pattern when the distributions are conditioned on recoil.

The $\Delta\chi^2$ in the lowest pT(Z) bin (< 10 GeV, $\Delta\chi^2 = 392$) is larger than in the 10–30 GeV bin. At very low pT(Z) the recoil effect is weak, so both families agree on the MET shape; however, the topology-mixture reweighting perturbs the event composition in this stratum even when the MET-scale effect is negligible, producing a larger discrepancy than in the bin where the two effects partially cancel.

The per-stratum MET distributions are shown in Fig. 3. In the three lower pT(Z) bins, Family A and Family B are visually close to the target. In the highest pT(Z) bin, Family B is systematically elevated relative to the target at low MET values and slightly suppressed at higher MET, while Family A tracks the target throughout.

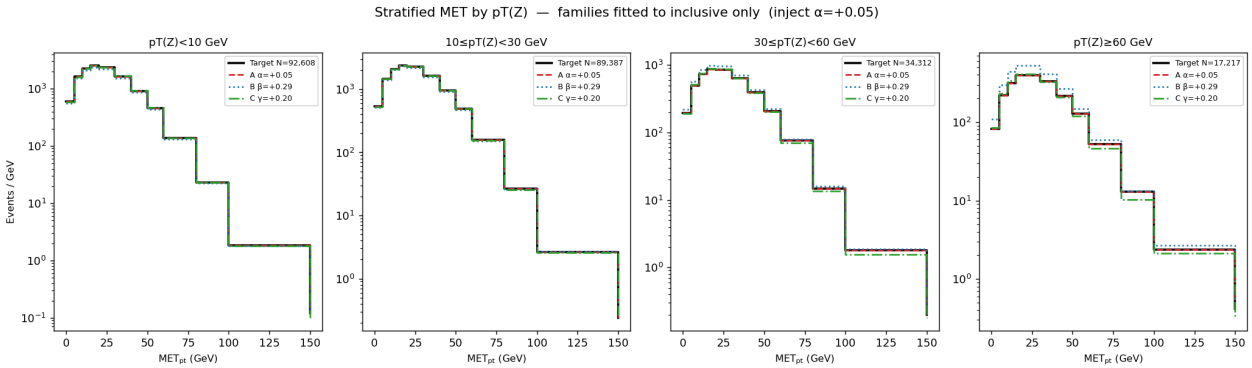

Figure 3: MET distributions in four pT(Z) strata: < 10 GeV (N = 92,608), 10–30 GeV (N = 89,387), 30–60 GeV (N = 34,312), and ≥60 GeV (N = 17,217). Target (black), Family A (dashed red), Family B (dotted blue), and Family C (dash-dot green) are shown at inclusive best-fit parameters without per-stratum refitting. The per-stratum discrimination $\Delta\chi^2(B) = \chi^2(B) - \chi^2(A)$ grows with recoil activity, reaching $\Delta\chi^2 = 1{,}432$ in the highest bin. Family C produces $\Delta\chi^2(C) = 58$ in the highest pT(Z) bin.

6.3 Jet-multiplicity stratification

As a complementary test, both models were evaluated after conditioning on the number of jets with pT > 30 GeV, cleaned against the selected Z muons using $\Delta R > 0.4$. The three jet categories follow the protocol binning: 0-jet, 1-jet, and ≥2-jet. Results are shown in Table 3 and Fig. 4.

Table 3. Per-stratum discrimination by jet multiplicity. $\Delta\chi^2 = \chi^2(B) - \chi^2(A)$ evaluated using best-fit parameters from the inclusive fit.

Jet bin	N	$\Delta\chi^2$
0-jet	171,385	1,405
1-jet	44,706	1,421
≥2-jet	17,433	3,415

Because the topology-mixture model modifies the relative weights of jet multiplicity categories globally, conditioning on jet count directly tests whether the model preserves the observed event population across topologies.

The jet-multiplicity stratification tests population consistency across event topologies rather than the MET–recoil relationship. Family B assigns global event weights of 1, $(1 + \beta)$, and $(1 + 2\beta)$ to the 0-jet, 1-jet, and ≥2-jet categories respectively, normalised to preserve the inclusive event count. At best-fit $\beta = +0.290$, the effective per-event weights after normalisation are approximately 0.91, 1.17, and 1.44 for the three categories. When the distribution is conditioned on jet multiplicity, this reweighting manifests as an inflated event count in the ≥2-jet stratum relative to the target, and a correspondingly depleted count in the 0-jet stratum.

Family A, by contrast, applies a continuous MET scale factor that does not alter the relative population of each jet stratum. The conditional event counts are therefore preserved by Family A and violated by Family B. The ≥2-jet bin produces the largest per-stratum $\Delta\chi^2$ (3,415), reflecting the 44% count inflation relative to the target. The discrimination in the 0-jet and 1-jet bins ($\Delta\chi^2 \approx 1{,}400$ each) reflects the corresponding count deficit and surplus in those categories.

The per-event $\Delta\chi^2$ ($\Delta\chi^2$ / N) is 0.0082, 0.032, and 0.196 for the 0-jet, 1-jet, and ≥2-jet bins respectively, confirming that the ≥2-jet stratum provides the strongest

population-level discrimination per event.

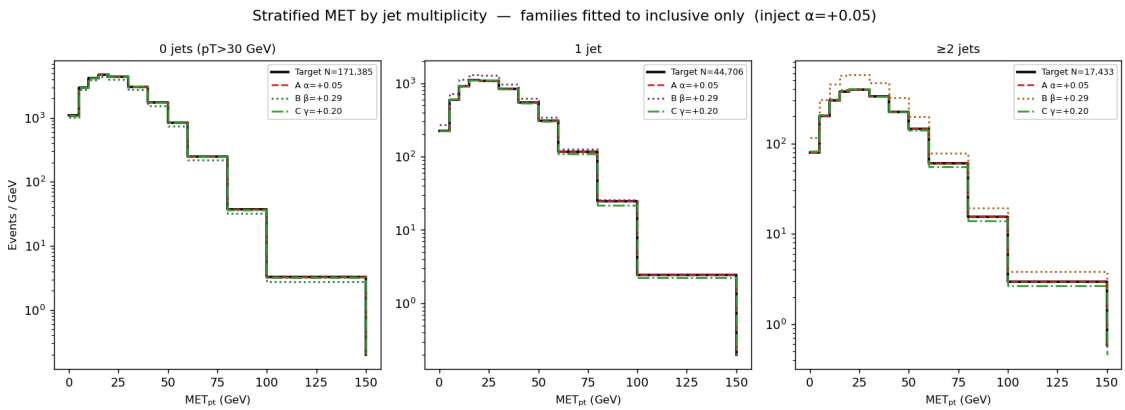

Figure 4: MET distributions in three jet-multiplicity categories: 0-jet (N = 171,385), 1-jet (N = 44,706), and ≥2-jet (N = 17,433). Target (black), Family A (dashed red), Family B (coloured dotted), and Family C (dash-dot green) are evaluated at inclusive best-fit parameters without refitting. Family B inflates the ≥2-jet event count by approximately 44% relative to the target, yielding $\Delta\chi^2(B) = 3{,}415$. Family C produces $\Delta\chi^2(C) \leq 26$ across all jet bins. Family A achieves $\chi^2(A) \approx 0$ throughout.

Together, Tables 2 and 3 show that both the recoil-stratified and jet-stratified projections expose clear discrepancies in Family B that are absent from the inclusive distribution. The two stratifications probe different latent assumptions: pT(Z) conditioning tests the MET–recoil relationship, while jet-multiplicity conditioning tests the population model.

6.4 Robustness to MET binning

The inclusive degeneracy demonstrated in Section 6.1 was evaluated under two alternative MET binning schemes in addition to the baseline (12 bins): a coarser binning with 9 bins merging adjacent low-MET intervals, and a tail-emphasised binning with modified boundaries above 50 GeV. In all three cases, $\chi^2(A)$ remained zero and the p-value for Family B exceeded 0.96. The inclusive degeneracy at the working point $\alpha = 0.05$ persisted under modest changes in MET binning, including coarser and tail-emphasised bin choices, with $p(B) > 0.96$ across all tested binning schemes. Detailed binning robustness results are reported in Supplementary Table S1.

Taken together, these tests demonstrate that while the inclusive MET distribution admits multiple nuisance explanations at modest distortion amplitudes, conditional projections restore discriminability by exposing violations of either recoil–response

structure or event-population consistency.

6.5 Physically grounded public-tier nuisance variation

To test whether the inclusive degeneracy demonstrated above extends beyond the phenomenological nuisance families introduced in Section 4, a third nuisance family is constructed based on detector-level systematic variations exposed directly in the CMS NanoAOD format.

NanoAOD provides event-level shifts corresponding to the CMS unclustered-energy systematic variation through the branches `MET_MetUnclustEnUpDeltaX` and `MET_MetUnclustEnUpDeltaY`. These quantities represent the change in the reconstructed missing transverse momentum vector when the unclustered-energy component of the event is shifted upward within the CMS reconstruction framework.

Family C — unclustered-energy vector perturbation — is defined by applying a scaled version of this shift to the nominal MET vector:

$$\mathrm{MET_C} = \mathrm{sqrt}((\mathrm{MET_x} + \gamma\ \Delta X)^2 + (\mathrm{MET_y} + \gamma\ \Delta Y)^2)$$

where MET_x = MET_pt × cos(φ), MET_y = MET_pt × sin(φ), and (ΔX, ΔY) are the NanoAOD unclustered-energy variation components. The parameter γ scales the magnitude of the systematic shift, with γ = 1 corresponding to the full unclustered-energy-up variation provided by CMS. The parameter search range is $\gamma \in [-3.0, +3.0]$ in steps of 0.05.

Fitting Family C to the inclusive target distribution produced by the recoil-response injection ($\alpha = 0.05$) yields a best-fit value $\gamma = +0.200$, corresponding to approximately 20% of the full CMS unclustered-energy shift. The resulting inclusive fit quality is $\chi^2(C) = 18.28$ (dof = 11, $p = 0.075$).

This value lies above the nominal rejection threshold ($p = 0.05$), indicating that the unclustered-energy perturbation is not rejected by the inclusive projection at this working point. However, the fit quality is substantially worse than that obtained for the topology-mixture family (Family B), which achieves $\chi^2 = 3.13$ and $p = 0.989$.

Conditional projections again expose discrepancies between the nuisance family and the injected recoil model. The recoil-stratified comparison produces $\Delta\chi^2(C)$ of {17, 5, 19, 58} across the four pT(Z) strata.

The contrast with the topology-mixture family is particularly pronounced in the highest recoil bin. In the pT(Z) ≥ 60 GeV stratum, the topology-mixture model produces $\Delta\chi^2(B) = 1{,}432$, whereas the unclustered-energy perturbation yields $\Delta\chi^2(C) = 58$. The ratio of these values (≈ 25) quantifies how much more efficiently the recoil stratification exposes the topology-mixture mechanism relative to the detector-level perturbation.

The jet-multiplicity stratification yields $\Delta\chi^2(C)$ of {11, 26, 24} for the 0-jet, 1-jet, and ≥2-jet categories respectively.

The smaller stratified discrepancies observed for Family C do not indicate that it provides a superior explanation of the target distribution. Rather, they reflect a difference in failure mode. The unclustered-energy perturbation modifies the MET vector while preserving the event-topology composition of the sample. As a result, jet-multiplicity conditioning does not directly expose a population inconsistency, in contrast to the topology-mixture family whose global reweighting violates the relative event counts across jet categories. The discriminating power of a given stratification variable therefore depends on which latent assumption of the nuisance model it probes.

These values are substantially smaller than those obtained for the topology-mixture family (Section 6.3), reflecting the fact that the unclustered-energy perturbation modifies the MET vector without altering the relative population of jet multiplicity categories. Consequently, the jet-stratified diagnostics expose Family C less strongly than Family B, which explicitly distorts the event-topology mixture.

Figure 1 shows the inclusive MET comparison including Family C alongside Families A and B. The unclustered-energy perturbation closely tracks the target distribution over most of the spectrum, but exhibits mild deviations in the intermediate MET region that account for the larger inclusive χ^2.

Taken together, these results indicate that physically grounded perturbations derived from CMS systematic variation branches can approach inclusive acceptability while remaining distinguishable through conditional projections. The strength of the degeneracy therefore depends on the underlying nuisance mechanism: topology-mixture distortions produce deep inclusive degeneracy, whereas detector-level perturbations such as unclustered-energy shifts exhibit weaker, partially recoverable degeneracy.

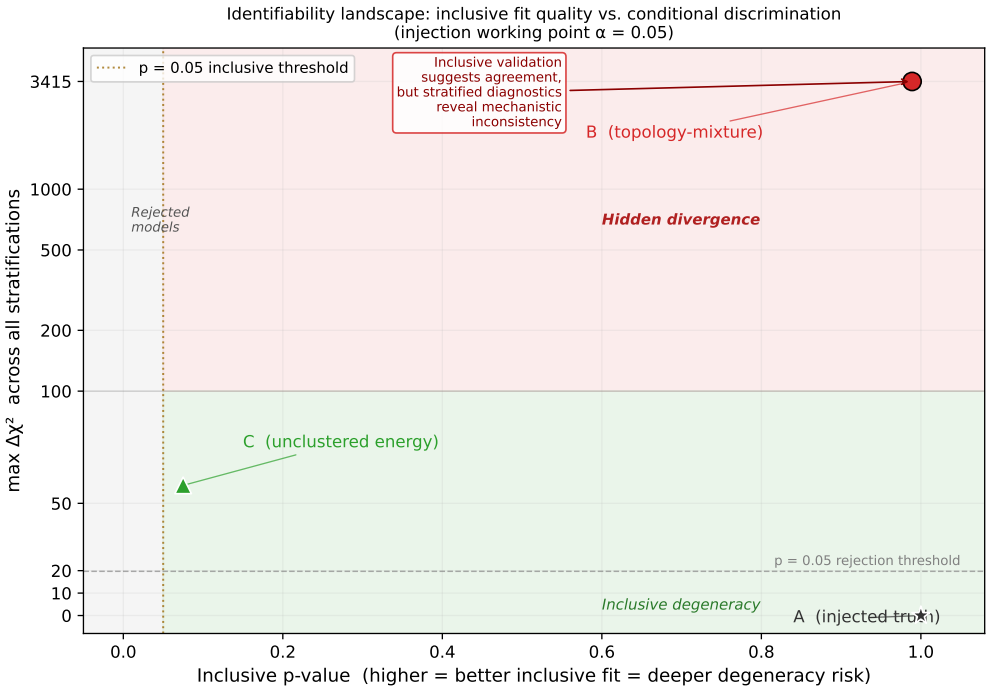

Figure 5: Identifiability landscape for nuisance mechanism families. The horizontal axis shows inclusive goodness-of-fit between simulated and observed MET distributions. The vertical axis shows the maximum conditional discrepancy across predefined strata. Configurations in the upper-right region exhibit strong inclusive agreement while producing large conditional discrepancies, illustrating how compression of collider events into reduced observables can mask mechanistic differences between nuisance models.

Table 4. Summary of identifiability across all three nuisance families at the working point $\alpha = 0.05$. Inclusive p-value and maximum per-stratum $\Delta\chi^2$ are reported for each family. A higher inclusive p-value indicates deeper degeneracy with the injected target (better inclusive fit); a higher max $\Delta\chi^2$ indicates stronger conditional exposure.

Family	Mechanism	Inclusive χ^2	p-value	max $\Delta\chi^2$
A	Injected truth	0.00	—	0
B	Topology-mixture	3.13	0.989	3,415
C	Unclustered-energy shift	18.28	0.075	58

Figure 5 summarizes the degeneracy structure of the nuisance families: several configurations that exhibit strong inclusive agreement with the observed MET distribution separate once conditional diagnostics are applied.

7. Discussion

7.1 Inclusive degeneracy and reduced observability

As demonstrated in Section 6, the inclusive MET distribution admits both nuisance families despite the large event sample. The topology-mixture model achieves $\chi^2(B) = 3.13$ on 11 degrees of freedom ($p = 0.989$), a fit quality statistically indistinguishable from the injected recoil-response model that generated the target distribution.

This result indicates a practical non-identifiability of nuisance explanations in the inclusive projection: the observed data are consistent with more than one generative model, and no hypothesis test on the inclusive projection can select among them. The non-identifiability here is not a consequence of insufficient data — 233,524 events is a large sample by typical analysis standards — but of insufficient observable dimensionality relative to the complexity of the nuisance space being probed.

The explanation lies in the nature of the inclusive projection. Both Family A and Family B alter the single-observable distribution P(MET) through mechanistically different operations. Family A applies a continuous MET scale factor that grows with pT(Z), shifting the MET distribution rightward in a recoil-correlated manner. Family B applies global per-event weights that shift the event-topology mixture, altering the effective composition of the sample without altering the MET scale within any given topology. At a sufficiently small distortion amplitude, the integrated shape changes produced by these two mechanisms can be made to agree to within statistical noise on the marginal distribution. The existence of a best-fit β that achieves this agreement — at a value of $\beta = +0.290$, well inside the search range — confirms that the degeneracy is not accidental but reflects the limited discriminating power of the inclusive one-dimensional projection.

This generalises a known challenge in MET analyses: inclusive validation distributes sensitivity across the entire observable space, potentially concentrating statistical power where signal-to-background from nuisance effects is weakest. The present study provides a quantitative illustration of this effect at a controlled and physically motivated working point. While the empirical example studied here involves missing transverse energy, the mechanism arises from observable compression and therefore applies to any collider analysis relying on reduced summary observables.

7.2 Stratified diagnostics as identifiability tests

The stratified analyses demonstrate that the non-identifiability of the inclusive distribution does not reflect a fundamental indistinguishability of the two nuisance models. Conditional projections P(MET | S) expose clear model differences that are invisible to the inclusive comparison P(MET).

The pT(Z) stratification tests the recoil–MET relationship. Family A is constructed so that its MET distortion is proportional to pT(Z): events with large pT(Z) receive large MET shifts, and events with pT(Z) ≈ 0 receive negligible shifts. When the analysis is conditioned on pT(Z), this design is directly tested — Family A must reproduce the target within each pT(Z) stratum using the same parameter fit to the inclusive distribution, and it does so with $\chi^2(A) \approx 0$ in all bins. Family B, which does not scale its distortion with pT(Z), cannot reproduce the conditional distributions while maintaining the parameter values that were optimised for the inclusive projection. The result is $\Delta\chi^2$ values ranging from 95 in the 10–30 GeV stratum to 1,432 in the ≥60 GeV stratum, a discrimination of three orders of magnitude relative to the inclusive $\chi^2(B) = 3.13$.

The jet-multiplicity stratification tests a different latent assumption: population consistency across event topologies. Family B assigns different event weights to the three jet categories, which globally alters the relative event counts within each category. When the distribution is conditioned on jet multiplicity, this population shift is directly exposed: Family B produces approximately 44% more events in the ≥2-jet stratum than the target, a count inflation that generates $\Delta\chi^2 = 3{,}415$ in that bin. Family A, which does not alter jet-category populations, achieves $\chi^2(A) \approx 0$ throughout.

The two stratifications therefore probe different latent assumptions of the alternative model: the pT(Z) test examines the recoil–MET relationship, while the jet-multiplicity test examines population consistency across event topologies. The pT(Z) test is sensitive to the functional dependence of MET on recoil activity; the jet test is sensitive to population consistency across topologies. A nuisance model that passes both conditioning tests would need to jointly preserve the recoil–MET relationship and the jet-category population fractions — a substantially more constrained requirement than agreement on the marginal MET distribution alone. The practical implication is that a two-dimensional stratification of this type can serve as a minimal identifiability test for inclusive nuisance estimates, without requiring access to additional observables

beyond those already provided at the NanoAOD tier.

Considering the three nuisance families together reveals that identifiability in reduced-observability analyses is graded rather than binary. The topology-mixture family produces a deep inclusive degeneracy that is strongly exposed by stratification, whereas the unclustered-energy perturbation occupies a boundary regime in which the inclusive projection is only marginally discriminating and conditional diagnostics produce moderate discrepancies. The efficiency with which stratification restores discrimination depends on which latent structural assumption of the nuisance family is violated.

7.3 Implications for analyses using compressed public datasets

Public analysis formats such as NanoAOD provide a reduced set of reconstructed observables and omit low-level detector quantities, effectively compressing the dimensionality of the observable event space. The CMS NanoAOD format in particular reduces per-event storage by retaining only high-level reconstructed quantities, discarding lower-level detector information. For external analyses working exclusively with public NanoAOD, the observable space available for nuisance characterisation is materially narrower than that available in a full AOD or MiniAOD analysis. This reduction has a direct implication for identifiability: as the dimensionality of the observable space decreases, the probability that P(X) is degenerate across distinct nuisance models increases for any single observable X.

The families studied here were chosen to be simple enough to fit and interpret with public data, but the degeneracy they exhibit is not a pathology of the specific choice. Any two nuisance models that produce compatible marginal distributions will be inclusive-indistinguishable on the compressed observable, regardless of their physical motivation. The analysis therefore points toward a systematic issue rather than an artifact: nuisance estimation from inclusive comparisons on compressed data may be subject to identifiability failures that remain invisible without explicit diagnostic checks.

Within collaboration analyses, Z→ll control samples are routinely used to constrain recoil-related nuisances in precision measurements such as the W boson mass [15]; the question addressed here is how much of that discriminating power remains available at the NanoAOD tier to an external analyst without full systematic-variation ac-

cess.

The conditional projections used here represent a minimal response to this issue, requiring only the stratification variables pT(Z) and jet multiplicity that are already present in NanoAOD. The approach requires no additional data access and imposes no significant computational overhead beyond the initial inclusive fit. It does require that the analyst commit to stratification variables in advance, before examining the stratified distributions, to avoid post-hoc selection of the stratification that best supports a preferred model — a standard requirement in confirmatory analysis that is made explicit in the frozen protocol accompanying this study.

7.4 Limitations and scope of the demonstration

Several limitations bound the conclusions that can be drawn from this study.

First, only two nuisance families are compared. The demonstration establishes that at least one topology-based explanation exists for the inclusive MET distribution at the working point $\alpha = 0.05$, but it does not characterise the full space of inclusive-indistinguishable models. A broader family survey would be required to map the degeneracy manifold and understand which model features are and are not constrained by the inclusive projection.

Second, the analysis is restricted to a single decay channel ($Z \to \mu^+\mu^-$) and a single observable (MET). The qualitative finding — that conditional projections restore discrimination absent from the marginal — is expected to hold more broadly, but the specific $\Delta\chi^2$ values and the working point at which degeneracy arises will depend on the channel, the nuisance families, and the observable.

Third, the nuisance families were constructed to be analytically tractable, not to represent the most realistic models of detector effects. Family A's linear recoil response and Family B's jet-count reweighting are simplifications of more complex physical processes. A study using more realistic nuisance families would require a more sophisticated fitting procedure and potentially a larger range of stratification variables to expose the relevant failure modes.

Fourth, the inclusive degeneracy demonstrated here is established at a specific working point ($\alpha = 0.05$, $\beta = +0.290$). The threshold sweep described in the supplementary material shows that inclusive discrimination is restored for $\alpha \geq 0.10$. The present re-

sult therefore characterises a regime of modest distortion amplitudes; whether similar degeneracies arise at other working points, or under different nuisance parameterisations, requires separate investigation.

These limitations notwithstanding, the central finding is robust: inclusive MET in $Z \to \mu^+\mu^-$ events from CMS Open Data admits a topology-mixture explanation at a physically plausible distortion amplitude, and two targeted conditional projections suffice to expose the failure of that explanation. The framework demonstrated here — inclusive fit, conditional evaluation, $\Delta\chi^2$ quantification — provides a practical template for identifiability diagnostics in compressed-data analyses. Within these limitations, the example provides a concrete demonstration that inclusive validation tests alone may be insufficient to guarantee interpretability in reduced-observability collider analyses.

A supplementary test using a physically grounded nuisance variation derived from CMS-exposed unclustered-energy systematic branches further clarifies the structure of the degeneracy. A scaled version of the unclustered-energy shift is not rejected by the inclusive test at the 5% level ($\chi^2 = 18.28$, 11 dof, $p = 0.075$), but produces noticeably poorer agreement than the topology-mixture model and yields only modest discrepancies under conditional stratification. This comparison suggests that non-identifiability in reduced-tier analyses is mechanism-dependent. Some nuisance families — particularly those altering event-topology mixtures — can produce deep inclusive degeneracies, whereas detector-level perturbations that preserve event topology remain more readily recoverable once conditional structure is examined.

The degeneracies demonstrated here arise from the restricted observable space of compressed public data tiers such as NanoAOD; they do not imply limitations in collaboration-internal analyses, which typically exploit higher-dimensional observables and systematic variation branches unavailable in the public format.

8. Conclusions

A total of 233,524 $Z \to \mu^+\mu^-$ candidate events were selected from CMS Open Data and used to compare two physically distinct nuisance families on the inclusive missing transverse momentum distribution. The topology-mixture family (Family B), fitted by grid search to the same target as the injected recoil-response family (Family A), achieves $\chi^2(B) = 3.13$ on 11 degrees of freedom ($p = 0.989$). The inclusive MET dis-

tribution does not reject the topology-mixture nuisance family at the working point $\alpha = 0.05$, despite more than 2×10^5 selected events.

Stratified analyses applied without refitting recover the discriminating information absent from the inclusive projection. Conditioning on pT(Z) yields $\Delta\chi^2 = \chi^2(B) - \chi^2(A) = 1{,}432$ in the highest pT(Z) bin (pT(Z) ≥ 60 GeV), with values growing monotonically with recoil activity. Conditioning on jet multiplicity yields $\Delta\chi^2 = 3{,}415$ in the ≥2-jet bin, reflecting the failure of the topology-mixture model to preserve the conditional event population across jet categories. The two stratifications probe different latent assumptions of the nuisance model — the recoil–MET relationship and population consistency across event topologies — and each exposes discrepancies absent from the inclusive view.

A supplementary test using the CMS unclustered-energy systematic variation shows that physically grounded perturbations can also approach inclusive degeneracy, though with weaker conditional discrepancies than the topology-mixture family. The comparison of three distinct nuisance families demonstrates that reduced-tier identifiability is not an all-or-nothing property but a spectrum determined by how the underlying mechanism projects onto the observable space.

The general principle illustrated is that inclusive validation constrains the marginal distribution P(MET), while stratified diagnostics probe conditional distributions P(MET | S), where S is a physics-motivated stratification variable. When analyses are performed on compressed data formats such as NanoAOD, the reduced dimensionality of the observable space increases the probability that P(MET) is degenerate across distinct nuisance models. Conditional projections provide a minimal mechanism to restore partial identifiability using the variables already available at the NanoAOD tier, without requiring access to lower-level detector information.

Identifiability diagnostics of the type demonstrated here should accompany inclusive validation tests in analyses that rely on inclusive nuisance estimation from compressed public datasets. The approach extends beyond the MET observable and the specific families studied: any inclusive comparison that constrains only P(X) for some observable X may be subject to similar degeneracies, and a targeted set of conditional projections P(X | S) provides a practical diagnostic framework. The present analysis is illustrative rather than exhaustive; a complete treatment would require a broader family of nuisance models and a systematic study of stratification variables. The central

finding — that inclusive MET in $Z \to \mu^+\mu^-$ events admits multiple nuisance explanations at physically plausible distortion amplitudes, while stratified projections distinguish them — demonstrates the interpretability risk in reduced-observability analyses and points toward a straightforward mitigation strategy. The use of real CMS Open Data events ensures that the observed degeneracy arises from properties of reduced collider observables themselves rather than artifacts of a particular simulation framework.

All analysis scripts and the frozen protocol used in this study are available in the accompanying repository, tagged at the version corresponding to this submission.

Data Availability

The CMS Open Data used in this analysis are publicly available through the CERN Open Data Portal (https://opendata.cern.ch). The specific dataset is the CMS 2016 DoubleMuon NanoAOD, accessible at the URL documented in Reference [1]. All analysis scripts, the frozen protocol document, and SHA-256 checksums for input data files are available in the accompanying repository at https://github.com/blumhardt-m/zmet-nid, tagged at the version corresponding to this submission.

AI Assistance Disclosure

Portions of the analysis code and manuscript text were developed with assistance from Claude (Anthropic) and Claude Code. All scientific decisions, experimental design, protocol definitions, and conclusions are the sole responsibility of the author. AI-assisted code was reviewed, tested, and validated by the author prior to use. The frozen protocol governing this analysis predates and is independent of any AI-assisted implementation steps.

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